METRIC

MIL-STD-1501G

22 Dec 2022

SUPERCEDING
MIL-STD-1501F
06 June 2011

DEPARTMENT OF DEFENSE
STANDARD PRACTICE

CHROMIUM PLATING, LOW EMBRITTLEMENT, ELECTRODEPOSITION

This document is activated for new design 22 Dec 2022

AMSC: NA

AREA: MFFP

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:31Z
Check the source to verify that this is the current version before use.

FOREWORD

1. This standard is approved for use by all Departments and Agencies of the Department of Defense.

2. This standard provides guidance repair process, acquisition, and manufacture of parts and/or spare parts on the landing gear of all military aircraft.

3. Comments, suggestions, or questions on this document should be addressed to: DAF OO-ALC/EN, Bldg. 849, Hill AFB, UT 84056, or emailed to OOALC.EN.Workflow@us.af.mil; DAF Landing Gear Program, Attn: Engineering at AFMC 417 SCMS/GUEA, Hill AFB, UT 84056-5228, or emailed to 417SCMS.GUEA.workflow@us.af.mil.. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:31Z
Check the source to verify that this is the current version before use.

CONTENTS

PARAGRAPH		PAGE
	FOREWORD	ii
	SUMMARY OF VERSION G MODIFICATIONS	v
1	SCOPE	1
1.1	Scope	1
1.2	Performance requirements.	1
1.3	Classification.	1
1.3.1	Classes.	1
1.3.2	Types.	
2	APPLICABLE DOCUMENTS	1
2.1	General	1
2.2	Government documents	1
2.2.1	Specifications, standards, and handbooks	1
2.2.2	Other Government documents, drawings, and publications	2
2.3	Non-Government publications	2
2.4	Order of Precedence	3
3	DEFINITIONS	3
3.1	High strength steel.	3
3.2	Material batch.	3
4	GENERAL REQUIREMENTS	3
4.1	General.	3
4.2	Materials and equipment.	3
4.2.1	Materials.	3
4.2.2	Equipment	3
4.3	Plated parts.	4
4.4	Finish.	4
4.5	Shot peening.	4
4.6	Embrittlement.	4
4.7	Reprocessing.	4
4.8	Plating thickness.	4
4.9	Class 1.	4
4.10	Class 2.	4

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:31Z
Check the source to verify that this is the current version before use.

CONTENTS (cont)

PARAGRAPH		PAGE
4.11	Class 3.	4
4.12	Type I plating.	4
4.13	Type II Plating.	5
5	DETAILED REQUIREMENTS	5
5.1	Individual tasks, requirements, or test methods.	5
5.1.1	Prior to plating.	5
5.1.2	Baking.	5
5.1.3	Plating sequence.	5
5.1.4	Storage of parts.	5
5.1.5	Handling of parts.	5
5.1.6	Masking.	5
5.1.7	Racking.	5
5.2	Plating procedure.	5
5.2.1	Step No. 1.	5
5.2.2	Step No. 2.	6
5.2.3	Step No. 3.	6
5.2.4	Step No. 4.	7
5.2.5	Step No. 5.	7
5.2.6	Step No. 6.	7
5.2.7	Step No. 7.	7
5.2.8	Lamination.	8
5.3	Post processing.	8
5.3.1	Grinding.	8
5.3.2	Penetrant inspection.	8
5.4	Process controls.	8
5.5	Qualification embrittlement test.	8
5.6	Process control embrittlement acceptance test.	9
5.7	Safety and health.	9
6	NOTES	9
6.1	Intended use.	9
6.2	Acquisition requirements.	9
6.3	Associated Data Item Descriptions (DID)	9
6.4	Subject term (key word) listing.	9
6.5	Changes from previous issue.	9
CONCLUDING MATERIAL		10

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:31Z
Check the source to verify that this is the current version before use.

SUMMARY OF VERSION G MODIFICATIONS

The following modifications to MIL-STD-1501G have been made:

PARAGRAPH			MODIFICATIONS
Page i	Title page		Updated
Page ii	Foreword		Updated
Page iii	Contents pages		Updated
Page iv	Change summary page		Added
1.2	Scope		Changed
1.2	Documentation		Changed
2.1	Government documents		Moved to 2.2 and changed to "General"
2.1.1	Specifications, standards, handbooks, and commercial item descriptions.		Moved to 2.2.1 and changed
2.2	Other government documents		Moved to 2.2.2 and changed
2.3	Non-Government publications.		Changed
4.1	General		Changed
4.1.1.c.	Anodes		Changed
4.2.2.a.	Power source		Changed
4.3	Specification		Changed to Plated parts
4.5	Shot peening		Changed
4.6	Embrittlement		Changed
4.10	Class 2		Changed
4.11	Class 3		Changed
4.13	Type II plating		Changed
5.1.5	Handling of parts		Changed
5.1.6	Masking		Changed
5.1.7	Racking		Changed
5.2.2	Step No. 2		Changed
5.2.3	Step No. 3		Changed
5.2.3.1			Changed
5.2.3.2			Added and changed
5.2.4	Step No. 4		Changed
5.2.5	Step No. 5		Changed
5.2.6	Step No. 6		Changed
5.3	Inspection		Deleted and changed to Post processing
5.2.8	Lamination		Added
5.2.8.1 - .3			Added
5.3	Post processing		Added
5.3.1	Grinding		Added
5.3.2	Penetrant inspection		Added
5.5	Qualification embrittlement test		Changed
5.5.a. – c.			Changed

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:31Z
Check the source to verify that this is the current version before use.

SUMMARY OF VERSION G MODIFICATIONS (cont.)

5.5.d.			Deleted
5.6.a. – b.			Changed
5.7	Safety and health		Changed
6.1	Intended use		Changed
6.2	Plating options		Changed
6.3	Subject term (key word) listing		Changed
6.4	Changes from previous issue		Deleted
	Concluding material		Updated

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:31Z
Check the source to verify that this is the current version before use.

1. SCOPE

1.1 Scope. This standard covers the detailed requirements for electrodeposition of hard chromium on high strength steel substrates and the required properties of the deposit. Subsequent heat treating techniques needed to insure low hydrogen embrittlement of steel are described.

1.2 Performance requirements. Chromium plated to this standard also meets the performance requirements of AMS 2460, which supersedes AMS-QQ-C-320. This standard may be used whenever AMS-QQ-C-320 or AMS 2460 is specified.

1.3 Classification

1.3.1 Classes. The following classes of chromium plating covered by this standard:

a. Class 1. Crack free (4.9)

b. Class 2. Limited cracking allowed (4.10)

c. Class 3. Moderate cracking allowed (4.11)

NOTE: When not specified, use Class 3 plating.

1.3.2 Types. The following types of chrome plating covered by this standard:

a. Type I. Plated to specified dimensions (4.12)

b. Type II. Processed to specific dimensions after plating (4.13)

NOTE: If not specified in the technical data, the processor may choose which type is appropriate based on the required thickness of the chromium deposit.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited within this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standard, and handbooks. The following specifications, standards, handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract. If the solicitation or contract does not specify an issue or revision, follow the current version of the applicable document(s).

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:31Z
Check the source to verify that this is the current version before use.

COMMERCIAL ITEM DESCRIPTIONS (CID)

A-A-55827	Chromium Trioxide, Technical
A-A-55828	Sulfuric Acid, Technical

(Copies of these documents are available online at https://assist.dla.mil)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-866	Grinding of Chrome Plated Steel and Steel Parts Heat Treated to 180,000 PSI or over
MIL-STD-871	Electro-Chemical Stripping of Inorganic Finishes
MIL-STD-1504	Abrasive Blast of Aircraft Components

(Copies of these documents are available online at https://assist.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

Title 29 CFR 1910 § Z	Occupational Safety and Health Standards (OSHA) – Toxic and Hazardous Substances

(Copies of these documents are available online at https://www.osha.gov/.

2.3 Non-Government publications. The following documents form a part of this standard to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM)

ASTM E 1444	Standard Practice for Magnetic Particle Testing
ASTM F 519	Standard Test Method for Mechanical Hydrogen Embrittlement Evaluation of Plating/Coating Processes and Service Environments

(Copies of these documents are available online at www.astm.org.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

AMS 2430	Shot Peening
AMS 2440	Inspection of Ground, Chromium Plated Steel Parts
AMS 2590	Rotary Flap Peening of Metal Parts
AMS 2644	Inspection Material, Penetrant

(Copies of these documents are available at https://www.sae.org)

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:31Z
Check the source to verify that this is the current version before use.

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this standard takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

3.1 High strength steel. For the purpose of this standard, high strength steel is defined as steel heat treated to 1.24 x 10 °Pa (180,000 psi) and above.

3.2 Material batch. All items processed at one time through the plating bath.

4. GENERAL REQUIREMENTS

4.1 General. Chrome plating is applied for wear resistance, hydraulic sealing surfaces, restoring dimensions for worn or undersized parts.

4.2 Materials and equipment.

4.2.1 Materials.

a. Chromium Trioxide, Technical (A-A-55827)

b. Sulfuric Acid, Technical (A-A-55828)

c. Anodes

Pure Lead.

Any Lead-Tin-Antimony alloy with a minimum lead composition of 93%.

Platinum coated metal.

4.2.2 Equipment.

a. Power source. Either generated or rectified direct current (D.C.) may be used. Ripple value shall not exceed 10 % (percent) as measured by dividing the Root Mean of Square of the alternating current (A.C.) voltage component by the D.C. voltage. This can best be measured by using an RMS A.C. voltage meter and dividing the value by the D.C. voltage. These measurements are to be taken across the anode and cathode bus at the tank.

b. Tanks. Tanks shall be resistant to the operating temperature and the chemical environment.

c. Temperature control. Plating tanks to be operated at temperatures other than room temperature shall be equipped with automatic temperature indicating and regulating devices.

d. Instrumentation. An ammeter shall be placed in series with the chromium tank cathode. The ammeter shall have sufficient shuts and switches to provide a full-scale reading equal to the maximum capacity of the power source and an accuracy of ± 5 % (percent) of the current being measured.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:31Z
Check the source to verify that this is the current version before use.

e. Blast equipment. The size of the cabinet shall be adequate to enclose the parts to be plated. Air lines shall be suitably trapped and filtered to remove moisture and prevent contamination of the part surface to be cleaned.

f. Bake oven. An oven capable of baking parts at 191 ± 14 °C (375 ± 25 °F) shall be located near the plating line. The size of the oven shall be adequate to enclose parts to be plated. The oven shall be equipped with temperature indicating, recording, and regulating devices.

4.3 Plated parts. All parts shall be plated in accordance with the requirements of AMS 2460 and in conjunction with those specified in this standard. If there is a conflict between these two documents, however, the requirements of this document shall govern. Adhesion testing shall be performed per AMS 2460 monthly in lieu of the test frequency required by AMS 2460.

4.4 Finish. The plated part shall have a finish that is continuous, homogenous, adherent, and free from harmful defects.

4.5 Shot peening. Unless otherwise specified, all parts shall be shot peened in accordance with AMS 2430 or AMS 2590, manual shot peening is allowed. Shot peening is not required on nickel plated surfaces. For Type 1 plating, the shot peened surface may need to be polished in order to achieve final surface finish.

4.6 Embrittlement. Qualification test specimens and process control test specimens shall be subjected to a sustained load test at 75% of the ultimate notched tensile strength for 200-hours minimum without failing or cracking. The plating/coating processes test method in accordance with ASTM F519.

4.7 Reprocessing. Parts rejected for defective plating, requiring stripping and re-plating, shall include all of the pre-plating steps of this standard. Parts shall be stripped in accordance with MIL-STD-871.

4.8 Plating thickness. The plating thickness shall be specified on the engineering drawing or other applicable directives. Except when otherwise specified, the minimum thickness shall be 0.05 millimeters (0.002 inches) and the maximum thickness .02 millimeters (0.008 inches) on the finished product.

4.9 Class 1. Parts plated to Class 1 shall show no mud cracking pattern when fluorescent penetrant inspected (5.3.1)

4.10 Class 2. Parts plated to Class 2 are allowed to show some cracking pattern when fluorescent penetrant inspected (see para 5.3.1). As a general rule, cracks exceeding 13 millimeters (1/2- inch) in length or individual areas of cracking exceeding 19 cm^2 (3 $inches^2$) or 10 % of plated areas, whichever is smaller, shall be cause for rejections of the plated area.

4.11 Class 3. Parts plated to Class 3 are allowed to show a moderate amount of cracking when fluorescent penetrant inspected (see para 5.3.2). To determine part acceptability, the part shall be cleaned to remove the penetrant materials and the areas of cracking inspected visually using an oblique light. If the cracking pattern is visible, that shall be cause for rejection of the plated area.

4.12 Type I plating. For Type I plating, the item shall be plated to the dimension and surface finish specified on the drawing. The surface finish on the item before plating shall be equal to or better than the required finish after plating. Type I plating may be buffed, polished, or lapped after plating if dimensional tolerance and surface conditions cannot be controlled in the plating operation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:31Z
Check the source to verify that this is the current version before use.

4.13 Type II plating. For Type II plating, a minimum of 0.05 millimeters (0.002 inches) more chrome than desired shall be deposited (per surface). The excess shall be ground off to give the final dimension and surface finish desired. Steel parts heat-treated to 1.24 x 109 Pa (180,000 psi) and above shall be ground in accordance with MIL-STD-866. All ground chrome plated surfaces shall be fluorescent penetrant inspected (see para 5.3.1). Chrome plated surfaces showing indications of abusive grinding (spiral, barber pole, circular, patch, or linear crack patterns), spalling or blistering shall be cause for rejection of the plated area, in accordance with AMS 2440. Chromium may be plated to size at thicknesses above 0.001" by employing anodes or other means of current control; in this case the chromium need not be ground although it may be honed or polished for surface finish.

5. DETAILED REQUIREMENTS

5.1 Individual tasks, requirements, or test methods.

5.1.1 Prior to plating. Except for finish grinding operations, all machining, forming, welding, and shot peening shall be completed prior to plating.

5.1.2 Baking. Parts shall be baked for stress relief before plating for four hours minimum at 191 ± 14 °C (375 ± 25 °F). Shot peening shall be performed before plating and after stress relieving.

5.1.3 Plating sequence. If chromium and cadmium are used in combination, the chromium shall be deposited first. When the chrome plating is to be followed by the cadmium plating, the 23-hour minimum bake period, required by para 5.2.7, can be replaced by four hours bake period at 191 ± 14 °C (375 ± 25 °F), provided the part is baked for 23-hours minimum after completion of the cadmium plating.

5.1.4 Storage of parts. Storage of parts between stress relief and cleaning shall be controlled to prevent contact with water or other corrosive materials. Parts shall be stored to permit free circulation of air around the parts.

5.1.5 Handling of parts. After the parts have been cleaned, they shall be handled in such a manner that shall minimize contamination of the surface to be plated.

5.1.6 Masking. Sections or areas of a part that are not to be plated shall be masked off. Masking materials that do not contaminate the plating bath shall be used. Masking shall be performed at the most convenient strep prior to plating.

5.1.7 Racking. Sufficient contact area and pressure shall be provide to carry the current without overheating the part. Racking shall be performed at the most convenient step prior to plating. When gang plating (two or more like parts) care shall be taken to assure a uniform division of current flow to each part by uniform take spacing and meticulous cleaning and shaping of racking contact surfaces or by providing isolated controlled current paths to each individual part; while maintaining the amperage limits specified (5.2.4).

5.2 Plating procedure. The chromium plating procedure shall be described below:

5.2.1 Step No. 1. Parts shall be vapor degreased, solvent cleaned or alkaline cleaned. Cathodic cleaning is not permitted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:31Z
Check the source to verify that this is the current version before use.

WARNING:
PROVIDE ADEQUATE VENTILIATION DURING DEGREASING OPERATIONS. AVOID SKIN AND EYE CONTACT WITH SOLVENT SOLUTIONS.

5.2.2 Step No. 2. The preferred method of cleaning is by dry blasting per MIL-STD-1504 using 80 to 320 grit aluminum oxide (A12O3), silicon dioxide (SiO2), or garnet. Hand sanding with soft backed abrasive materials is also allowed. Other non-embrittling cleaning processes may be used if they result in satisfactory adhesion when tested per the requirements of this standard; grinding is also an allowed verification of satisfactory adhesion. Elapsed time between completion of Step No. 2 and Step No. 3 shall not exceed four (4) hours.

5.2.3 Step No. 3. Anodic etch 15.5 to 46.5 amps/decimeter^2 (1 to 3 amps/inch^2) for 30-seconds to 10-minutes. Etch in the following solution (preferred) or in the plating bath:

Chromic Acid: 225 to 300 gram/liter (30 to 40 ounce/gallon)
Temperature: 32 to 60 °C (90 to 140 °F)

CAUTION: Etching increases iron contamination in the plating bath; therefore, the use of the above solution is recommended.

5.2.3.1 Alternate reverse etch solutions. A sulfuric acid or a sulfuric-hydrofluoric acid mixture can be used as an alternate, reverse etch solution to the solution in accordance with 5.2.3.

5.2.3.2 The conditions for the use of either the sulfuric acid or sulfuric-hydrofluoric acid mixture are as follows:

a. 40% sulfuric acid:

(1) Operating range – Concentrated (93%) sulfuric acid diluted to 35 to 45% by volume.

(2) Temperature – Ambient

(3) Current density – 31.0 to 62.0 amps/decimeter^2 (2 to 4 amps/inch^2) anodic.

(4) Time – 30 to 120 seconds

b. Sulfuric-Hydrofluoric Acid:

(1) Operating range – Concentrated (93%) sulfuric acid diluted to 20 to 30% by volume and concentrated (70%) hydrofluoric acid diluted to two (2) to seven (7) % by volume. Ammonium bifluoride may be used at an equivalent total fluoride concentration in place of hydrofluoric acid.

(2) Temperature – Ambient

(3) Current density – 31.0 to 62.0 amps/decimeter^2 (2 to 4 amps/inch^2) anodic.

(4) Time – 60 to 90 seconds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:31Z
Check the source to verify that this is the current version before use.

c. 25% Sulfuric Acid:

(1) Operating range – Concentrated (93%) sulfuric acid diluted to 20 to 30% sulfuric acid by volume.

(2) Temperature – Ambient.

(3) Current density – 31.0 to 62.0 amps/decimeter^2 (2 to 4 amps/inch^2) anodic.

(4) Time – 30 to 120 seconds.

5.2.4 Step No. 4. Chromium plate at 15.5 to 77.5 amps/decimeter^2 (1 to 5 amps/inch^2), to the required thickness. Chromium plate in the following situations:

a. Chromic Acid – 225 to 270 g/l (30 to 36 ounce/gallon)

b. Sulfuric Acid – 2.25 to 2.7 g/l (0.30 to 0.36 ounce/gallon)

c. Trivalent Chrome – 4 g/l maximum

d. Iron – 5 g/l maximum

e. Ratio Chromic Acid/Sulfuric Acid – 80/1 to 120/1

f. Temperature – 52 to 62 °C (125 to 145 °F)

g. It is acceptable to start plating at a higher current (two to three times the normal plating current) for a short time (approximately 30 to 60 seconds), then reduce to the desired plating current. This may help in providing the initial layer of chromium and lowers the chance of bare spots in the chrome deposit.

NOTE: Other plating solutions may be used when approved by the Responsible Engineering Organization provided the deposit meets the requirements of this standard.

WARNING:
PROVIDE ADEQUATE VENTILATION WHEN USING CHROMIC PLATING SOLUTIONS. WEAR PROTECTIVE RUBBER GLOVES AND CHEMICAL GOOGLES TO PREVENT SKIN AND EYE EXPOSURE.

5.2.5 Step No. 5. Rinse parts in cold water and inspect for defects and adequate buildup. If undersized, return to plating tank and continue plating. To restart plating, follow the chrome lamination steps in paragraphs 5.2.8. If plating is adequate, remove masking, racking, and clean as necessary.

5.2.6 Step No. 6. Rinse all parts in hot water and blow dry with compressed air. Elapsed time between completion of plating and start of baking shall not exceed four hours.

5.2.7 Step No. 7. Bake all parts heat treated to 1.24 x 109 PA (180,000 psi) and above 23-hours minimum at 191 ± 14 °C (375 ± 25 °F) (see para 5.1.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:31Z
Check the source to verify that this is the current version before use.

5.2.8 Lamination (optional). Chrome may be laminated onto used chrome surfaces, provided that the original chrome surface is sound and shows no signs of subsurface corrosion, poor adhesion, or other quality defects. The following steps must be used to ensure proper adhesion.

Clean the existing chrome surface using hand methods, alkaline or vapor degreasing; or a combination of these three methods, **do not blast**. Mask surfaces that are not to be plated, at any convenient step.

Immerse the component in the plating solution and allow the part to come to temperature. Reverse etch the chrome surface at one to three amps per square inch for 30 to 120 seconds.

Reduce the current flow to zero, then switch the rectifier or connections to forward polarity. Slowly ramp the current from zero up to the required plating current over the course of five to ten minutes; plate until desired thickness is achieved.

5.3 Post processing. Inspection shall be in accordance with the inspection and tests in AMS 2460 and this standard.

5.3.1 Grinding. Grinding of the chrome, when required, shall be in accordance with MIL-STD-866.

5.3.2 Penetrant inspection. If the chrome is ground, penetrant inspect the ground surface in accordance with MIL-STD-866 with pass/fail criteria in accordance with AMS 2440.

5.4 Process controls. Solutions used in the plating process shall be checked periodically and maintained in accordance with the requirements of this process standard.

5.5 Qualification embrittlement test. The vendor shall demonstrate their ability to provide chromium plate which passes sustained load testing at 75% of the ultimate notched tensile strength for 200-hours without failure using four notched 4340 steel Type 1a.1 or 1a.2 specimens in accordance with ASTM F519.

a. The specimens shall be prepared for plating and plated in accordance with all of the requirements of this standard and ASTM F519 section 10.1. During plating the specimens shall be mounted symmetrically on a rack by themselves. All areas of the rack except the contact area shall be coated with a suitable maskant. An ammeter, having a sensitivity of 0.5 amperes or better shall be connected in the plating circuit near the specimens being plated. The ammeter shall either be built into the rectifier being used or shall be inserted between the rectifier and the cathode. The entire surface between the grip areas shall be plated with chromium. The minimum chrome thickness shall be 0.002” when measured on the cylindrical (non-notched) section of the specimen. The specimens shall be plated at two to three amperes per square inch. The specimens shall be baked for 23-hours at 191 ± 14 °C (375 ± 25 °F) within four (4) hours of removal from the bath.

b. The test shall be considered passing if all four specimens last 200-hours at 75% ultimate notched tensile strength without failure.

c. Upon successful completion of the static load test, one of the notched tensile specimens shall be sectioned across the notch parallel to the axis of the specimen. Photomicrographs shall be taken of the notched area and examined for complete chromium coverage of the notch (use 80 to 100 X magnification).

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:31Z
Check the source to verify that this is the current version before use.

5.6 Process control embrittlement acceptance test. The process control embrittlement acceptance test shall be as follows:

a. Two (2) standard specimens of the type noted in para 5.5 shall be plated in accordance with paragraph 5.2 at two (2) to three (3) amps per square inch to a minimum thickness of 0.002" as measured on the cylindrical (non-notched) section of the specimen. The entire surface between the grip areas shall be coated with chrome. The specimens shall be subjected to a sustained load test of 75% of the ultimate notch tensile strength for 200-hours minimum and in accordance with the requirements of 4.6. Failure of any one of the specimens shall constitute failure of the test, production shall cease until the cause of the failure is determined, and the bath requalified. Acceptance of the items completed after the last successfully completed acceptance test shall be withheld until the extent and cause of the failure has been determined.

b. The test for embrittlement shall be conducted as often as deemed necessary with a maximum interval of 90-days. Prior to processing with a new bath, the bath must be qualified in accordance with 5.5.

5.7 Safety and health. This document specifies the use of certain materials which have been listed in 29 Code of Federal Regulations (CFR) 1910 Occupational Safety and Health Administration (OSHA) Standards as "Toxic and Hazardous Substances". Personnel exposed to these materials during the process must be limited to the values specified in applicable portions of the 29 CFR 1910.1000. While the materials, methods, applications, and processes described or referenced in this specification may involve the use of hazardous materials, this specification does not address the hazards which may be involved in such use. It is the responsibility of the user to ensure familiarity with the safe and proper use of any hazardous materials and to take necessary precautionary measures to ensure the health and safety of all personnel involved.

6. NOTES

This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.

6.1 Intended use. Chrome plating is applied for wear resistance, reconditioning worn or undersized parts, and such incidental corrosion protection as the specified thickness of the deposit may afford. This standard provides guidance on the procedures and processes for low hydrogen embrittlement chromium electrodeposition on high strength steel components; however, the processes described within this document, may also be used to deposit chromium on low strength steels, stainless steels, and nickel surfaces. Where corrosion protection is desired, an undercoat of nickel 0.025 to 0.050 millimeters (0.001 to 0.002 inches) thickness is recommended. When not specified, use the preferred Class 3 plating (1.3.1).

6.2 Subject term (key word) listing.
Chrome
Corrosion
Process Control

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:31Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army – MR
Navy – SH
Air Force – 70

Preparing Activity
Air Force – 70

(Project MFFP-2022-001)

Review Activities:
Navy – AS
Air Force – 20

NOTE: The activities listed above were interested in this standard as of the date of this standard. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:31Z
Check the source to verify that this is the current version before use.